C - Fishbones

分值：300 分

Problem Statement

Artist Takasago has created an object in the shape of a fish skeleton.

艺术家高砂创作了一个鱼骨架形状的艺术品。

The object consists of N ribs and one spine. The ribs are numbered 1 through N.

该艺术品由 N 根肋骨和一根脊椎组成。肋骨编号为 1 到 N。

He wants to write one string on each of the $N+1$ bones, satisfying all of the following conditions.

他希望在 $N+1$ 根骨头上各写一个字符串，满足以下所有条件。

- The length of the string written on the spine is N.
- For each rib $i = 1, \ldots, N$, the following hold.
  - The length of the string written on rib i is A_i.
  - The B_i-th character of the string written on rib i equals the i-th character of the string written on the spine.
- Each of the strings written on the $N+1$ bones is one of $S_1, \cdots, S_M$ (duplicates allowed).
- 写在脊椎上的字符串长度为 N。
- 对每根肋骨 $i = 1, \ldots, N$，满足以下要求：
  - 写在肋骨 i 上的字符串长度为 A_i。
  - 肋骨 i 上字符串的第 B_i 个字符与脊椎上字符串的第 i 个字符相同。
- 写在 $N+1$ 根骨头上的每个字符串都属于 $S_1, \cdots, S_M$（允许重复）。

$S_1, \cdots, S_M$ are strings consisting of lowercase English letters, and they are all distinct.

$S_1, \cdots, S_M$ 是由小写英文字母组成的字符串，且两两互不相同。

For each $j = 1, \cdots, M$, answer the following question.

对每个 $j = 1, \cdots, M$，回答以下问题。

- Among the ways to write strings satisfying the conditions, is there one where the string written on the spine is S_j?
- 在所有满足条件的字符串书写方案中，是否存在一种方案使得脊椎上的字符串为 S_j?

Constraints

- N is an integer.
- $1 \le N \le 10$
- A_i and B_i are integers. $(1 \le i \le N)$
- $1 \le B_i \le A_i \le 10$ $(1 \le i \le N)$
- M is an integer.
- $1 \le M \le 200\,000$
- S_j is a string consisting of lowercase English letters. $(1 \le j \le M)$
- $1 \le |S_j| \le 10$ $(1 \le j \le M)$
- $S_1, \cdots, S_M$ are pairwise distinct.
- N 是一个整数。
- $1 \le N \le 10$
- A_i 和 B_i 是整数。 $(1 \le i \le N)$
- $1 \le B_i \le A_i \le 10$ $(1 \le i \le N)$
- M 是一个整数。
- $1 \le M \le 200\,000$
- S_j 是由小写英文字母组成的字符串。 $(1 \le j \le M)$
- $1 \le |S_j| \le 10$ $(1 \le j \le M)$
- $S_1, \cdots, S_M$ 两两互不相同。

Input

The input is given from Standard Input in the following format:

输入按以下格式从标准输入给出：

N
$A_1\ B_1$
$\vdots$
$A_N\ B_N$
M
S_1
$\vdots$
S_M

Output

Output M lines.

输出 M 行。

The j-th line ($1 \leq j \leq M$) should contain `Yes` if there exists a way to write strings satisfying the conditions with S_j written on the spine, and `No` otherwise.

第 j 行（$1 \leq j \leq M$）中，如果存在满足条件的书写方案使得脊椎上写的是 S_j，则输出 `Yes`，否则输出 `No`。

Sample Input 1

```
5
5 3
5 2
4 1
5 1
3 2
8
retro
chris
itchy
tuna
```

```
crab
rock
cod
ash
```

Sample Output 1

```
Yes
Yes
No
No
No
No
No
No
```

By writing `chris`, `retro`, `tuna`, `retro`, `cod` on ribs $1, 2, 3, 4, 5$ respectively, the conditions are satisfied with `retro` written on the spine.

在肋骨 $1, 2, 3, 4, 5$ 上分别写下 `chris`、`retro`、`tuna`、`retro`、`cod`，即可满足条件，此时脊椎上写的是 `retro`。

- The length of `retro` is 5.
- For each rib, the following hold.
  - The string written on rib 1 is `chris`, which has length 5. Its third character is `r`, which equals the first character of `retro`.
  - The string written on rib 2 is `retro`, which has length 5. Its second character is `e`, which equals the second character of `retro`.
  - The string written on rib 3 is `tuna`, which has length 4. Its first character is `t`, which equals the third character of `retro`.
  - The string written on rib 4 is `retro`, which has length 5. Its first character is `r`, which equals the fourth character of `retro`.
  - The string written on rib 5 is `cod`, which has length 3. Its second character is `o`, which equals the fifth character of `retro`.

- retro 的长度为 5。
- 对每根肋骨，满足以下要求：
    - 肋骨 1 上写的字符串是 chris，长度为 5。它的第三个字符是 r，与 retro 的第一个字符相同。
    - 肋骨 2 上写的字符串是 retro，长度为 5。它的第二个字符是 e，与 retro 的第二个字符相同。
    - 肋骨 3 上写的字符串是 tuna，长度为 4。它的第一个字符是 t，与 retro 的第三个字符相同。
    - 肋骨 4 上写的字符串是 retro，长度为 5。它的第一个字符是 r，与 retro 的第四个字符相同。
    - 肋骨 5 上写的字符串是 cod，长度为 3。它的第二个字符是 o，与 retro 的第五个字符相同。

By writing itchy, chris, rock, itchy, ash on ribs $1, 2, 3, 4, 5$ respectively, the conditions are satisfied with chris written on the spine.

在肋骨 $1, 2, 3, 4, 5$ 上分别写下 itchy、chris、rock、itchy、ash，即可满足条件，此时脊椎上写的是 chris。

Sample Input 2

```
5
5 1
5 2
5 3
5 4
5 5
8
retro
chris
itchy
tuna
crab
rock
```

```
cod
ash
```

Sample Output 2

```
Yes
Yes
Yes
No
No
No
No
No
```